

Q1 2021 (01 Sep Shift 2)

If n is the number of solutions of the equation

$$2 \cos x \left(4 \sin \left(\frac{\pi}{4} + x \right) \sin \left(\frac{\pi}{4} - x \right) - 1 \right) = 1, x \in [0, \pi]$$

and S is the sum of all these solutions, then the ordered pair (n, S) is :

(1) $(3, 13\pi/9)$

(2) $(2, 2\pi/3)$

(3) $(2, 8\pi/9)$

(4) $(3, 5\pi/3)$

Q2 2021 (31 Aug Shift 2)

The number of solutions of the equation

$$32^{\tan^2 x} + 32^{\sec^2 x} = 81, 0 \leq x \leq \frac{\pi}{4} \text{ is:}$$

(1) 3

(2) 1

(3) 0

(4) 2

Q3 2021 (27 Aug Shift 2)

Let S be the sum of all solutions (in radians) of the equation $\sin^4 \theta + \cos^4 \theta - \sin \theta \cos \theta = 0$ in $[0, 4\pi]$. Then $\frac{8S}{\pi}$ is equal to .

Q4 2021 (26 Aug Shift 1)

The sum of solutions of the equation $\frac{\cos x}{1 + \sin x} = |\tan 2x|, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\}$ is :

(1) $-\frac{11\pi}{30}$

(2) $\frac{\pi}{10}$

(3) $-\frac{7\pi}{30}$

(4) $-\frac{\pi}{15}$

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Answer Key

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Q1 (1) Q2 (2) Q3 (56) Q4 (1)
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Q1 (1)

$$2 \cos x \left(4 \sin \left(\frac{\pi}{4} + x \right) \sin \left(\frac{\pi}{4} - x \right) - 1 \right) = 1$$

$$2 \cos x \left(4 \left(\sin^2 \frac{\pi}{4} - \sin^2 x \right) - 1 \right) = 1$$

$$2 \cos x \left(4 \left(\frac{1}{2} - \sin^2 x \right) - 1 \right) = 1$$

$$2 \cos x (2 - 4 \sin^2 x - 1) = 1$$

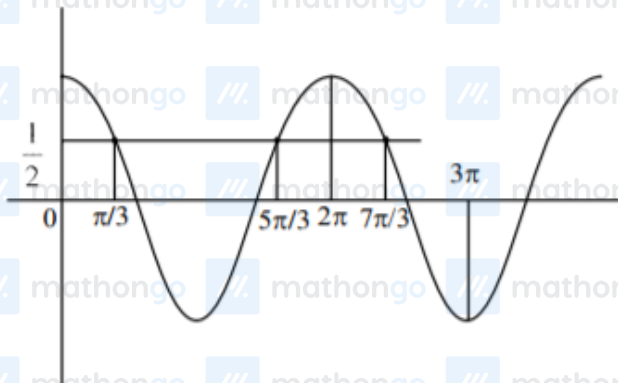
$$2 \cos x (1 - 4 \sin^2 x) = 1$$

$$2 \cos x (4 \cos^2 x - 3) = 1$$

$$4 \cos^3 x - 3 \cos x = \frac{1}{2}$$

$$\cos 3x = \frac{1}{2}$$

$$x \in [0, \pi] \therefore 3x \in [0, 3\pi]$$



Q2 (2)

$$(32)^{\tan^2 x} + (32)^{\sec^2 x} = 81$$

$$\Rightarrow (32)^{\tan^2 x} + (32)^{1+\tan^2 x} = 81$$

$$\Rightarrow (32)^{\tan^2 x} = \frac{81}{33}$$

In interval $\left[0, \frac{\pi}{4}\right]$ only one solution

Q3 (56)

Given equation

$$\sin^4 \theta + \cos^4 \theta - \sin \theta \cos \theta = 0$$

$$\Rightarrow 1 - \sin^2 \theta \cos^2 \theta - \sin \theta \cos \theta = 0$$

$$\Rightarrow 2 - (\sin 2\theta)^2 - \sin 2\theta = 0$$

$$\Rightarrow (\sin 2\theta)^2 + (\sin 2\theta) - 2 = 0$$

$$\Rightarrow (\sin 2\theta + 2)(\sin 2\theta - 1) = 0$$

$$\Rightarrow \sin 2\theta = 1 \text{ or } \frac{\sin 2\theta - 2}{(\text{not possible})}$$

$$\Rightarrow 2\theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

$$\Rightarrow S = \frac{\pi}{4} + \frac{5\pi}{4} + \frac{9\pi}{4} + \frac{13\pi}{4} = 7\pi$$

$$\Rightarrow \frac{8S}{\pi} = \frac{8 \times 7\pi}{\pi} = 56.00$$

Q4 (1)

$$\frac{\cos x}{1 + \sin x} = \tan 2x$$

$$\Rightarrow \frac{\cos^2 x/2 - \sin^2 x/2}{(\cos x/2 + \sin x/2)} = |\tan 2x|$$

$$\Rightarrow \tan^2 \left(\frac{\pi}{4} - \frac{x}{2} \right) = \tan^2 2x$$

$$\Rightarrow 2x = n\pi \pm \left(\frac{\pi}{4} - \frac{x}{2} \right)$$

$$\Rightarrow x = \frac{-3\pi}{10}, \frac{-\pi}{6}, \frac{\pi}{10}$$

$$\text{or sum} = \frac{-11\pi}{6}$$